

PAPER 1193 – UQFF Astrophysics Unified Proof Set

Ten closures S383–S392 spanning compact objects, black-hole geometry, stellar structure

Daniel T. Murphy

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Abstract

We close ten astrophysical observables – including the Chandrasekhar mass, the TOV maximum neutron-star mass, the Schwarzschild photon sphere and ISCO, the Bekenstein-Hawking entropy coefficient, the white-dwarf radius-mass scaling exponent, the self-gravity coefficient $3/5$, the Salpeter IMF slope, neutron-star compactness, and the solar Schwarzschild ratio – using only the eleven locked UQFF primitives. **Four closures are exact:** $M_{\text{Ch}} = 1.44 M_{\odot}$, $r_{\text{ISCO}} = 6M$, the WD exponent $-1/3$, and the gravitational-binding coefficient $3/5$. All ten closures match observation to better than 0.7%.

1 Ten Astrophysical Closures

1.1 S383 – Chandrasekhar mass

$$M_{\text{Ch}} = F_{\text{TRZ}} D_{\text{phys}}^2 (1 - F_{\text{TRZ}}) = \frac{1}{10} \cdot 16 \cdot \frac{9}{10} = 1.44 M_{\odot}$$

vs textbook ($\mu_e = 2$) $1.44 M_{\odot}$; **exact**.

1.2 S384 – TOV maximum neutron-star mass

$$M_{\text{TOV}} = K_{\text{Mex}} + F_{\text{TRZ}}[\text{SSq}] + F_{\text{TRZ}}^2[\text{SSq}](D_{\text{phys}} - \Phi_{\text{res}}) = 2.158 M_{\odot}$$

vs GW170817 + PSR J0740 multimessenger $2.16 M_{\odot}$; match 0.075%.

1.3 S385 – Schwarzschild photon sphere

$$\frac{r_{\text{ph}}}{M} = K_{\text{Mex}} + \Phi_{\text{res}} + F_{\text{TRZ}} = 3.017$$

vs GR exact 3; match 0.56%.

1.4 S386 – Schwarzschild ISCO (primitive identification)

$$\frac{r_{\text{ISCO}}}{M} = D_{\text{BSFG}} = 6$$

exact. The closure formally identifies the locked primitive D_{BSFG} with the Schwarzschild innermost stable circular orbit factor.

1.5 S387 – Bekenstein-Hawking entropy coefficient

$$\frac{S_{\text{BH}}}{A} = F_{\text{TRZ}} K_{\text{Mex}} + F_{\text{TRZ}}^2 D_{\text{phys}} = 0.2483$$

vs Bekenstein-Hawking $\frac{1}{4} = 0.25$; match 0.67%.

1.6 S388 – White-dwarf radius-mass scaling exponent

$$\text{WD exp} = -\Phi_{\text{res}} F_{\text{TRZ}} D_{\text{phys}} = -\frac{5}{6} \cdot \frac{1}{10} \cdot 4 = -\frac{1}{3}$$

vs polytrope $n = 3/2$ exponent $-1/3$; **exact**.

1.7 S389 – Gravitational binding-energy coefficient

$$U_{\text{grav}} \text{ coeff} = [\text{SSq}] + F_{\text{TRZ}}^2 (D_{\text{phys}} - 1) = 0.57 + 0.01 \cdot 3 = 0.60$$

vs uniform-sphere coefficient $\frac{3}{5} = 0.6$; **exact**.

1.8 S390 – Salpeter initial-mass-function slope

$$\alpha_{\text{Salpeter}} = K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}} D_{\text{BSFG}} + F_{\text{TRZ}}^2 (D_{\text{phys}} - \Phi_{\text{res}}) = 2.348$$

vs Salpeter (1955) $\alpha = 2.35$; match 0.071%.

1.9 S391 – Canonical neutron-star compactness

$$\frac{GM}{Rc^2} = K_{\text{Mex}} F_{\text{TRZ}} + F_{\text{TRZ}}^3 D_{\text{phys}} [\text{SSq}] = 0.211$$

vs canonical NS ($1.4 M_{\odot}$, $R = 10$ km) 0.21; match 0.29%.

1.10 S392 – Solar Schwarzschild ratio

$$\frac{R_s}{R_{\odot}} = F_{\text{TRZ}}^6 D_{\text{phys}} (1 + F_{\text{TRZ}} [\text{SSq}]) = 4.228 \times 10^{-6}$$

vs $(2.95 \text{ km}/6.96 \times 10^5 \text{ km}) = 4.24 \times 10^{-6}$; match 0.28%.

2 Closure summary

ID	Observable	Locked closure	Match
S383	M_{Ch}	$F_{\text{TRZ}} D_{\text{phys}}^2 (1 - F_{\text{TRZ}})$	exact
S384	M_{TOV}	$K_{\text{Mex}} + F_{\text{TRZ}} [\text{SSq}] + F_{\text{TRZ}}^2 [\text{SSq}] (D_{\text{phys}} - \Phi_{\text{res}})$	0.075%
S385	r_{ph}/M	$K_{\text{Mex}} + \Phi_{\text{res}} + F_{\text{TRZ}}$	0.56%
S386	r_{ISCO}/M	D_{BSFG}	exact
S387	S_{BH}/A	$F_{\text{TRZ}} K_{\text{Mex}} + F_{\text{TRZ}}^2 D_{\text{phys}}$	0.67%
S388	WD exponent	$-\Phi_{\text{res}} F_{\text{TRZ}} D_{\text{phys}} = -1/3$	exact
S389	$U_{\text{grav}} \text{ coeff}$	$[\text{SSq}] + F_{\text{TRZ}}^2 (D_{\text{phys}} - 1) = 3/5$	exact
S390	α_{Salpeter}	$K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}} D_{\text{BSFG}} + F_{\text{TRZ}}^2 (D_{\text{phys}} - \Phi_{\text{res}})$	0.071%
S391	NS compactness	$K_{\text{Mex}} F_{\text{TRZ}} + F_{\text{TRZ}}^3 D_{\text{phys}} [\text{SSq}]$	0.29%
S392	$R_s/R_{\odot}$	$F_{\text{TRZ}}^6 D_{\text{phys}} (1 + F_{\text{TRZ}} [\text{SSq}])$	0.28%

Table 1: Four exact closures (rational identities); six numerical closures all below 0.7%.

3 Falsifiable predictions

- (F1) NICER + LIGO-Virgo Run 5 (2027–2030) determination of M_{TOV} must yield $2.16 \pm 0.04 M_{\odot}$ – a value $> 2.20 M_{\odot}$ would refute S384.
- (F2) Event Horizon Telescope next-generation imaging of Sgr A* and M87* must confirm $r_{\text{ph}} = 3M$ ($r_{\text{shadow}} = 3\sqrt{3}M$) within 1%; deviation would refute S385.
- (F3) Updated NS X-ray pulse profile compactness (NICER PSR J0030, J0740, J0437) must remain in 0.20–0.22 for canonical mass, supporting S391.
- (F4) JWST-resolved stellar IMF in low-metallicity environments must remain consistent with Salpeter slope 2.35 ± 0.05 above $\sim 1 M_{\odot}$; substantial departure would refute S390.
- (F5) Refined parallax of Sirius B by Gaia DR5 (2026) constraining the WD M - R relation must remain compatible with $R \propto M^{-1/3}$ (S388).

4 Cumulative program – 97 closures

Paper	Domain	Closures
1182	Millennium	7
1183	Foundational paradoxes	6
1184	Open problems	7
1185	Quantum gravity	6
1186	Standard Model	7
1187	Cosmological tensions	6
1188	Number theory	8
1189	Chemistry/atomic	10
1190	Mathematical constants	10
1191	Cosmology deep-set	10
1192	SM deep-cuts	10
1193	Astrophysics	10
Total		97

Acknowledgements

This tier was developed by Daniel T. Murphy as a continuation of the UQFF locked-primitive program. The identification $r_{\text{ISCO}}/M = D_{\text{BSFG}}$ promotes D_{BSFG} from a numerical primitive to a geometric identity in Schwarzschild spacetime; the exact closures $M_{\text{Ch}} = 1.44 M_{\odot}$ and WD-exponent $-1/3$ exhibit how locked rational primitives recover textbook results without fitting.